LAB 10

Prepare a Software Quality Management Plan • Lab duration: 105 minutes

IIIT Vadodara | IT367 Software Project Management | Autumn 2026-27

A. Group and Project Details

Field	Student input
Project title	Abhinay – Film Production Networking and Casting Platform
Group number / team	Group 1
Team members and roll numbers	Ishan Agrawal (202452321), Happy (202452319), Abhishek (202452339), Arjun S (202452308, client representative), Triya (202452338), Priyadarshini (202452324), Himanshu (202452320, Quality Manager), Kshithesh (202452329), Yuvaraju (202452325), Nikitha (202452336)
Project manager / coordinator	Ishan Agrawal
Submission date	26/10/2026
Project scope or major features	(1) Multi-role registration, secure login and professional profiles with portfolio, CV, photos and reels; (2) Casting marketplace: role posting, search, apply, shortlist folders and application status tracking (Applied / Shortlisted / Selected / Rejected); (3) Direct messaging, scheduled auditions, self-tape upload, project pages and admin moderation.

B. Quality Objectives and Measurable Targets (4-6 required)

ID	Quality objective	Metric / how to measure	Target	Responsible role
Q1	Every high-priority requirement is covered by tests	Traceability matrix: high-priority FRs with at least one test case ÷ total high-priority FRs	100% of high-priority FRs and ≥ 80% of medium-priority FRs have a test case	Quality Manager
Q2	Authentication and role-based access are secure	Role-based access test cases passed; unauthorised access to private applications/messages that succeeds (Lab 9 risk R4)	100% of access-control tests pass; 0 unauthorised-access defects open at delivery	Backend Developer / Quality Manager
Q3	Few defects remain at delivery	Open defects by priority from the defect log at the final integration build	0 open High-priority defects; ≤ 5 open Low-priority defects	Quality Manager
Q4	Frontend and backend work together reliably	API/integration tests passed against the agreed API contracts and validation rules (Lab 9 risk R2)	≥ 95% of API/integration tests pass in the final build	Full-stack Team
Q5	System responds quickly in normal use	Page/API response time measured in browser dev tools with about 50 concurrent users; chat message delivery delay	Main pages and search results load in ≤ 3 seconds; chat messages delivered in ≤ 1 second	Backend Developer
Q6	Code is clean and maintainable	Pull requests reviewed before merge; lint errors reported by the linter on the main branch	100% of merged PRs peer-reviewed; 0 lint errors on main	Configuration & Change Manager

C. Quality Assurance (QA) Plan - Prevent Problems

QA activity	When will it happen?	Evidence to keep	Responsible role
Requirement review	Before design – end of Week 3 (Lab 6 plan)	Signed-off review checklist and approved requirement list (Lab 2)	Scope & Requirements Manager
Code style review / peer review of every pull request	Before each merge to main, Weeks 4–11	Pull request review comments and linter report	Configuration & Change Manager
API contract and data-model review	Before each module build starts (Weeks 4–11)	Agreed API specification and review notes	Full-stack Team
Test-case review and traceability check	Before integration testing, Weeks 11–12	Test cases mapped to FR/NFR IDs in the traceability matrix	Quality Manager

D. Quality Control (QC) Plan - Find Problems

QC activity / test	Feature or deliverable	Pass criterion	Responsible role
Login, registration and role-based access testing	User Registration & Authentication (1.1.1)	Valid users log in; invalid credentials rejected; users cannot open another user's or role's private data	Backend Developer / Tester
Input validation testing	Profile, casting role posting and application forms	Empty, wrong-format or oversized inputs and files are rejected with a clear message; valid input is saved	Frontend Developer / Tester
Application workflow and status testing	Role Discovery & Application, Application Status Tracking (1.2.2, 1.3.2)	Applicant can apply once; status moves only to valid states; applicant is notified on each change	Full-stack Team / Tester
Messaging and self-tape upload testing	Direct Messaging and Self-Tape Submission (1.4.1, 1.4.3)	Messages delivered in real time; videos upload and play back; unsupported file types are blocked	Backend Developer / Tester

E. Defect Recording and Closure Process

Defect ID	Problem / expected vs actual	Priority	Assigned to	Status	Retest outcome
D01	Example: Any logged-in user can change an application status by calling the status API directly. Expected: only the producer/director who owns the posting can update it.	High	Backend Developer	Closed	Pass
D02	Example: Profile form accepts an oversized reel file without any error. Expected: file over the size limit is rejected with a message.	Medium	Frontend Developer	Fixed	Pending

Closure rule (when do we call a defect closed?): A defect is closed only when the fix is merged after peer review, the tester re-runs the failing test case and it passes (with related regression tests), the evidence (screenshot/test log) is saved in the defect log, and the Quality Manager marks it Closed. Flow: identify → report in log (ID, description, priority) → assign → fix → retest → close, or reopen if the retest fails.

F. Acceptance Criteria for Major Deliverables

An acceptance criterion tells how you will decide whether a feature or final output is ready to accept.

Deliverable	Acceptance criterion (clear and testable)	Evidence	Who approves?
Requirements document	All high- and medium-priority FRs and NFRs reviewed, with no unresolved review comments	Requirement review record and signed-off document	Client representative (Arjun S) and Project Manager
Casting marketplace (role posting, search, apply)	Producer can publish, edit and close a posting; actor can search and apply once; all related test cases pass	Test results and demo	Client representative and Quality Manager
Application tracking and messaging	Status changes update correctly and notify the applicant; messages are delivered in real time	Test results and demo	Client representative and Quality Manager
Final software (Abhinay web application)	All in-scope features work in the deployed build; Q1–Q6 targets met; 0 open High defects; documentation delivered	Final test report, defect log summary and demo of the deployed build	Client representative and Project Manager

G. Assumptions, Reflection and Final Check

Assumptions or data sources (from previous labs): Group details, scope and requirements are from Labs 1–2, the WBS from Lab 3, size and effort from Labs 4–5 (4.75 KLOC), the schedule from Labs 6–7 and risks R2, R4, R8 from Lab 9. Quality targets are planning values assumed by the team. The defect entries are examples, as no real defects have been logged yet.

Reflection by project manager (2-3 sentences): What did you learn about the quality of your project? How will this plan guide your next activity?

Our main quality concerns are access control, API mismatches between team members and defects found late, which match the High risks from Lab 9. This plan lets the Quality Manager track measurable targets from the start, so reviews and tests happen before the integration phase in Weeks 11–14 instead of at the very end.

Submission checklist: ☑ 4-6 measurable quality objectives ☑ QA plan ☑ QC plan ☑ assigned roles ☑ defect closure process ☑ major acceptance criteria ☑ consistent with previous labs